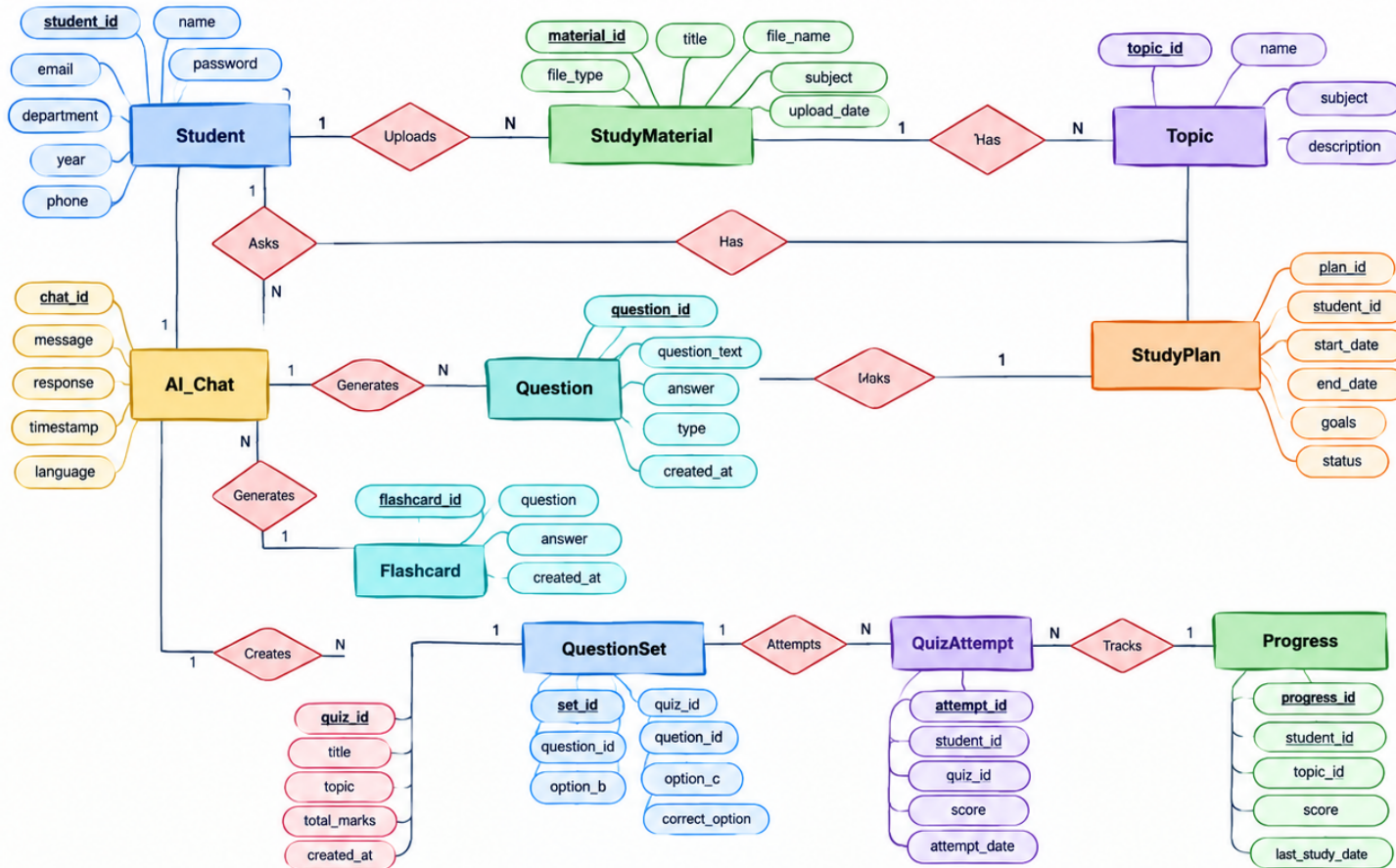


AI StudyBuddy – ER Diagram

Entity Relationship Diagram for the AI-powered personalized learning assistant

AI StudyBuddy – ER Diagram

An Intelligent Personalized Learning Assistant for Students



Legend

- Entity
- ◆ Relationship
- Attribute
- 1 — N Cardinality (One to Many)
- underline Primary Key
- (FK) Foreign Key

Entities and Key Attributes

1. Student (student_id) – PK
2. StudyMaterial (material_id) – PK
3. Topic (topic_id) – PK
4. AI_Chat (chat_id) – PK
5. Question (question_id) – PK
6. Quiz (quiz_id) – PK
7. QuestionSet (set_id) – PK
8. QuizAttempt (attempt_id) – PK
9. Progress (progress_id) – PK
10. StudyPlan (plan_id) – PK
11. Flashcard (flashcard_id) – PK

Relationships Summary

1. Student uploads StudyMaterial (1:N)
2. StudyMaterial has Topic (1:N)
3. Student asks Question (1:N)
4. AI_Chat generates Question (1:N)
5. Question belongs to Topic (N:1)
6. Student has StudyPlan (1:N)
7. Student creates Quiz (1:N)
8. Quiz contains QuestionSet (1:N)
9. QuestionSet linked to Question (N:1)
10. QuizAttempt made by Student (N:1)
11. QuizAttempt tracks Progress (N:1)
12. AI_Chat generates Flashcard (1:N)